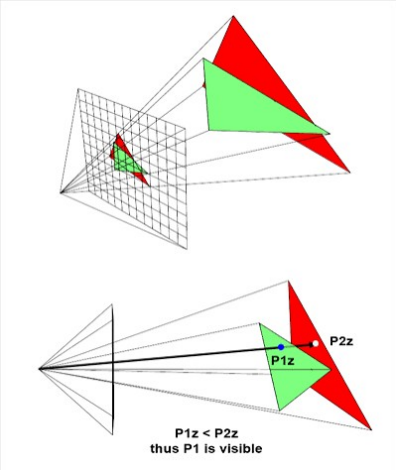

Raytracing Workshop
Keller
3-230
10/24
5:45pm
gopher.graphics/events

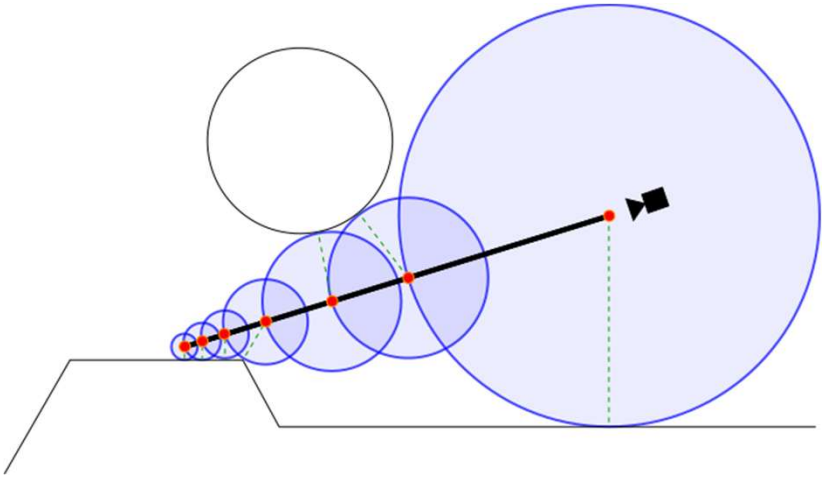
Motivation

- How do we render a 3D scene to a 2D image?

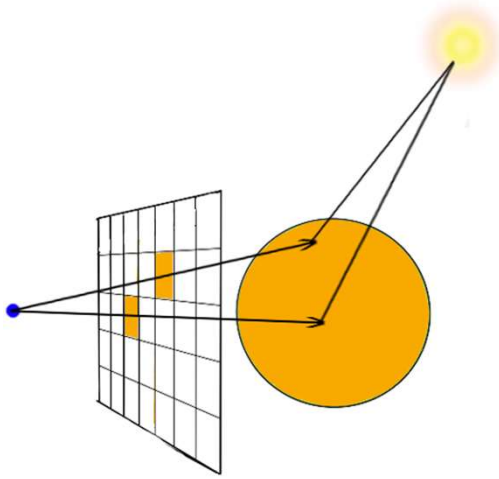
Rasterization



Ray Marching



Raytracing



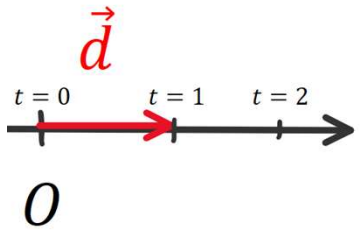
How do we simulate light?

- Treat individual photons as a ray
- Rays come from light sources with an intensity and a color
- Rays bounce around the scene and “collect” color
- Rays hit the camera sensor
- Record the pixel color where the light ray hit the camera
- Realistic, but very slow



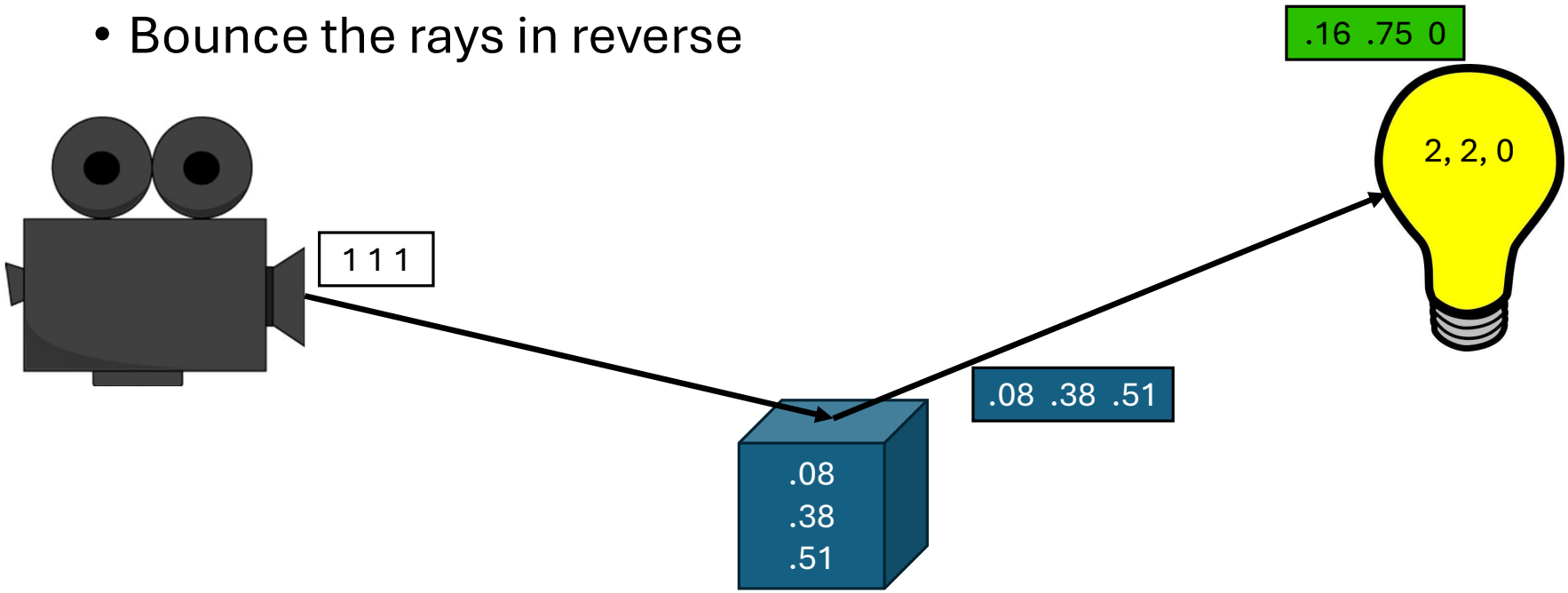
How do we represent rays?

- With an origin O and a non-zero direction $\vec{d}$
- Rays start at the origin and extend infinitely in their direction
- A *positive* time value t gives a position along the ray
- $R(t) = O + t\vec{d}$



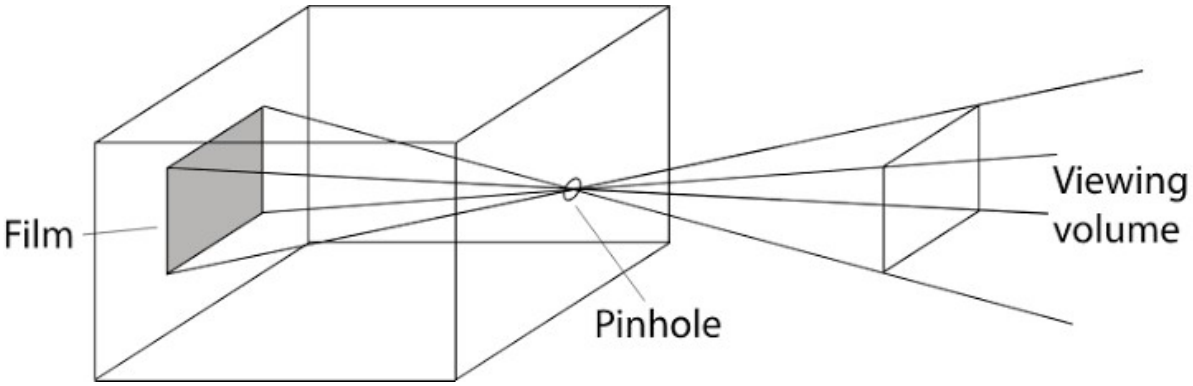
Make it faster

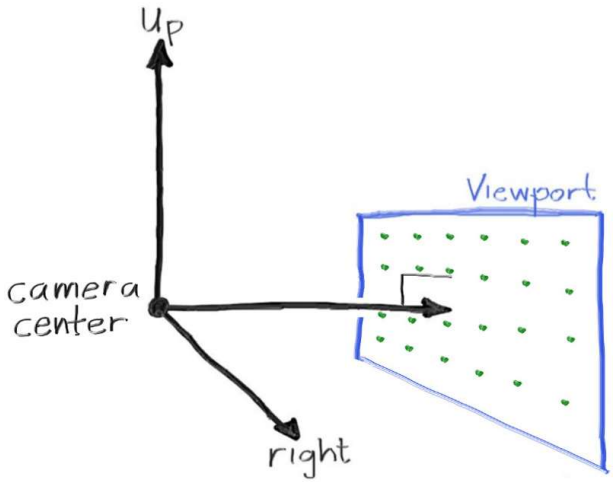
- Multiplying RGB components approximates “collecting” light
- Multiplication is communitive
- Bounce the rays in reverse



How do we send rays?

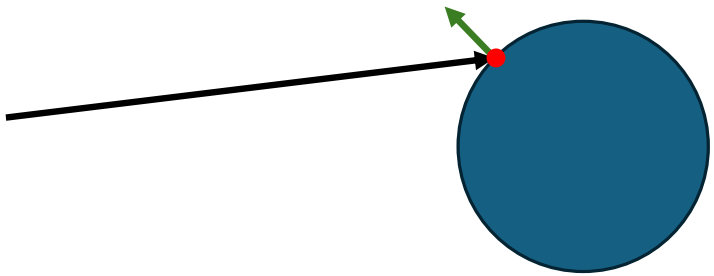
- Limit the angles each pixel can accept
- Rays start at the camera center and travel out through the viewport





How do rays bounce?

- Unlike ray marching, rays do not “move forward”
- We find the intersection points analytically
- Different formula for each shape in the scene
- Produces the time and point of intersection
- Produces the normal vector at that point



Ray – Sphere intersection

$$\|(P - C)\|^2 = r^2$$

$$\left\|\left(O + t\vec{d}\right) - C\right\|^2 = r^2$$

$$\left(t\vec{d} + (O - C)\right) \cdot \left(t\vec{d} + (O - C)\right) = r^2$$

$$t^2\left(\vec{d} \cdot \vec{d}\right) + 2t\vec{d} \cdot (O - C) + (O - C) \cdot (O - C) = r^2$$

$$t^2\left(\vec{d} \cdot \vec{d}\right) + t\left(2\vec{d} \cdot (O - C)\right) + \left((O - C) \cdot (O - C) - r^2\right) = 0$$

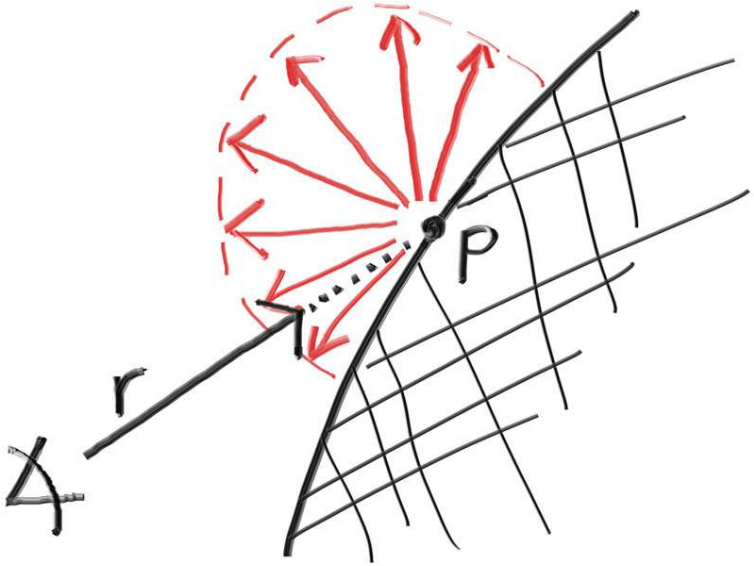
$$a = \left(\vec{d} \cdot \vec{d}\right), b = 2\vec{d} \cdot (O - C), c = (O - C) \cdot (O - C) - r^2$$

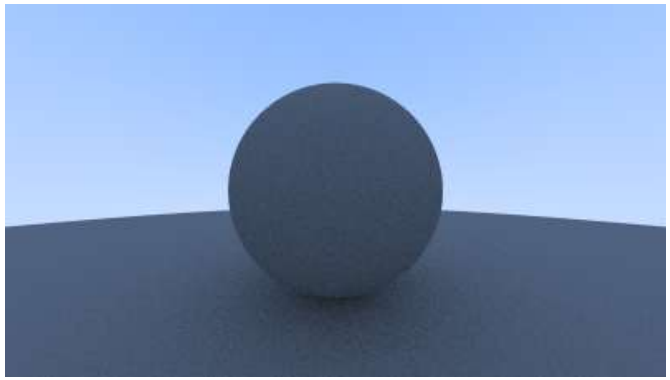
Ray scattering

- After a ray hits a surface, we need to scatter it
- Influences the material properties of the surface
- Diffuse scattering
- Lambertian scattering
- Reflective scattering
- Dielectric scattering (next week)

Diffuse scattering

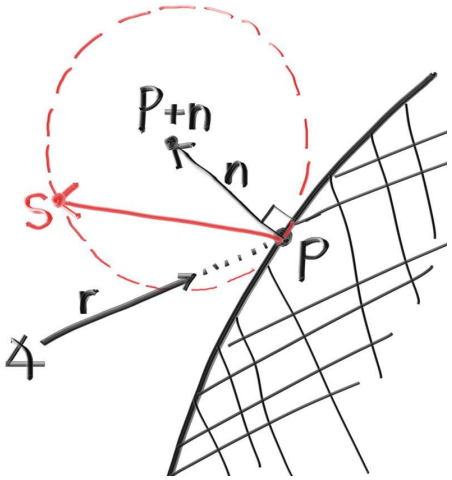
- Mimics rough, non-reflective surfaces
- A ray is bounced in a random direction in the same hemisphere as the normal

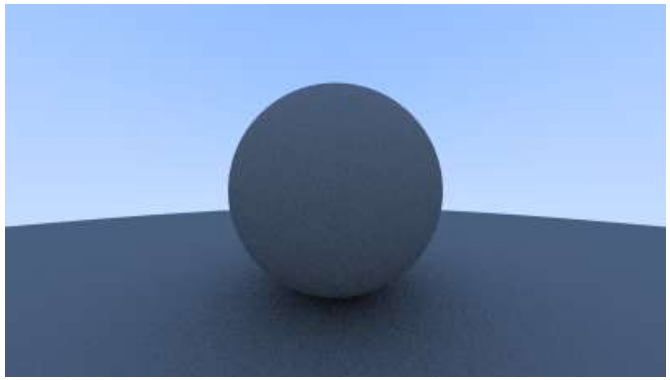




Lambertian scattering

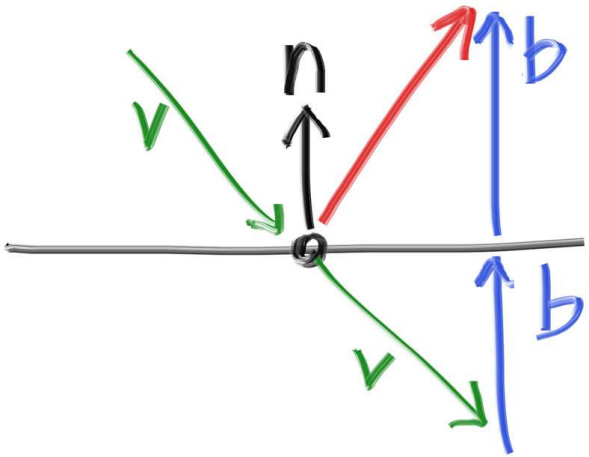
- Similar to diffuse scattering
- Rays are more likely to scatter near the normal

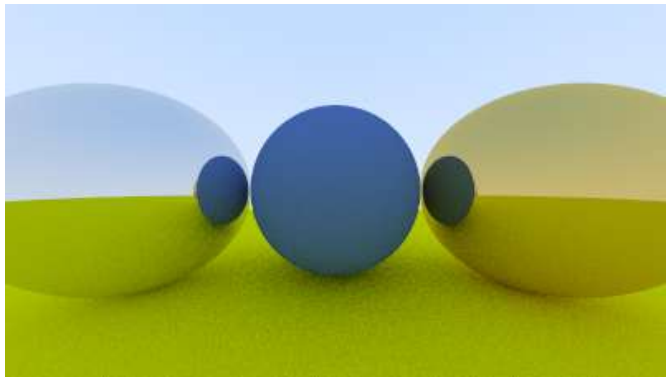




Reflective scattering

- Used to mimic metals, mirrors, and other reflective surfaces





Questions about ray tracing

The secret super cool raytracing website